

6.5.1 Using X-rays

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) basic structure of an X-ray tube; components – heater (cathode), anode, target metal and high voltage supply	

- (b)** production of X-ray photons from an X-ray tube
- (c)** X-ray attenuation mechanisms; simple scatter, photoelectric effect, Compton effect and pair production
- (d)** attenuation of X-rays; $I = I_0 e^{-\mu x}$, where μ is the attenuation (absorption) coefficient *M0.5, M3.11*
- (e)** X-ray imaging with contrast media; barium and iodine HSW9, 10, 12
- (f)** computerised axial tomography (CAT) scanning; components – rotating X-tube producing a thin fan-shaped X-ray beam, ring of detectors, computer software and display
- (g)** advantages of a CAT scan over an X-ray image. HSW9, 10, 12